

DBA LAB

Subject Code	CP08524P
Semester	5
Hours per Semester	60 h
Revision Number	0
Revision Date	22.08.2022

Course Outcomes:

After completing this subject, the trainees will be able to:

1. Explain installation and configuration of SQL Server.
2. Set up the database in SQL Server.
3. Explain how to manage security in SQL Server.
4. List different backup and recovery methods.
5. Observe the use of automation using SQL Agent.
6. Discuss database Maintenance and query tuning.

List of Exercises:

Topic No.	Topics / Subtopics	Allotted Hours
1	Introduction	1 h
2	Installation of SQL Server 2012	2 h
	2.1 Lab Exercise : Step by Step Installation guide for SQL Server 2012	2
3	SQL Server Management Studio (SSMS)	2 h
	3.1 Lab Exercise - Exploring the SSMS	1
	3.2 Lab Exercise - SSMS Configurations, Tips and Issues	1
4	Architecture	2 h
	4.1 SQL Server QUERY TREE Viewer (optional demo)	1
	4.2 Lab Exercise: Viewing a QUERY PLAN	1
5	Setting up DB	4 h
	5.1 Lab Exercises to CREATE, ALTER AND DELETE User Databases	4

6	Security Management	12 h
	6.1 Demo : Security Components	4
	6.2 Lab Exercise : Create Login, User, assign Permission	5
	6.3 SQL Server Data Encryption	1.5
	6.4 Lab Exercise : Row Level Security (RLS)	1.5
7	Backup and Recovery	12 h
	7.1 Lab Exercise: Types of recovery models	1
	7.2 Understanding SQL Server Backup Types	2
	7.3 Backup Commands T-SQL Recap	3
	7.4 Create backup using SQL Server Management Studio SSMS	3
	7.5 SQL Server Restore Options and Commands Tutorial	3
8	SQL Server Agent	6 h
	8.1 Demo of Jobs and Schedules	5
	8.2 Analyzing Problems with SQL Server Agents	1
9	Project 1 : Security	3 h
10	DB Maintenance	6 h
	10.1 Lab Exercise : Maintenance Plan Wizard - Backup Schedule	3
	10.2 Lab Exercise: Page Level Restore for a Corrupted database using DBCC	3
11	Performance Tuning	10 h
	11.1 Example of Indexing Tuning	1
	11.2 Tools and Utilities: DMV – Dynamic Management Views	1
	11.3 Tools and Utilities: DB Engine Tuning Advisor	2
	11.4 Tools and Utilities: Profiler	2
	11.5 Tools and Utilities : Performance Monitor or PerfMon	2
	11.6 Deadlocks	2
	Total	60 h